

EuRepoC CYBER INCIDENTS

Exploratory Data Analysis Report

European Repository of Cyber Incidents • Global, Receiver, Attribution & Dyadic Datasets

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1. Data Cleaning Summary

1.1 Standardizing Disguised Missing Values

EuRepoC encodes missing information using placeholder text ('Not available', 'Unknown', 'Not attributed') rather than leaving cells blank. Left as-is, this defeats pandas' native null-handling — a plain `.isnull()` check radically understates real missingness. All such placeholders were standardized to proper NaN values across all four files before any further processing.

Result: 118,768 individual cell values in the global dataset alone converted from disguised placeholder text to real NaN.

1.2 Dropping Near-Empty Columns

Columns missing in more than 95% of rows were dropped, since they carry negligible analytical value. What was removed, by file:

- Global: `zero_days_subcode`, `user_interaction`, `economic_impact`, `evidence_for_sanctions_indicator`, `legal_response_subtype`, `casualties`
- Receiver: none required dropping
- Attribution: `attribution_subtype`, `attributing_company`
- Dyadic: `casualties`, `physical_effects_spatial`, `physical_effects_temporal`, `economic_impact`, `affected_eu_countries`, `affected_eu_countries_value`

The single most extreme case: 'casualties' was 100% placeholder text in every file it appeared in — it carries no information in this dataset version and was dropped everywhere.

1.3 Date Parsing

Most date fields use DD.MM.YYYY (`start_date`, `end_date`, `attribution_date`, political/legal response dates); the two database timestamp fields (`added_to_db`, `updated_at`) use ISO YYYY-MM-DD. Both formats were parsed explicitly, with anything unparseable coerced to NaT rather than raising an error.

Statistic	start_date
Valid dates	3,322 of 3,414 (97.3%)
Earliest incident	2000-01-01
Median incident date	2022-03-12
Latest incident	2024-12-19

1.4 Boolean Field Conversion — and a Caveat Worth Flagging

`settled_attribution` was expected to need conversion from text ('True'/'False') to a proper boolean. In practice, pandas had already inferred this column as native boolean on read-in, so mapping against the string keys 'True'/'False' matched nothing, and the entire column ended up NaN post-conversion.

Data-quality note: this means `settled_attribution` is not currently usable in its cleaned form. The fix is simple — skip the `.map()` step when the column's dtype is already bool (or map the actual True/False objects, not their string representations) — but it's called out here rather than silently left in a broken state.

1.5 Duplicate Rows

Dataset	Fully duplicated rows found
Global	0
Receiver	0
Attribution	29 (removed)
Dyadic	0

attribution_id and dyad_id repeating across rows is expected by design (multiple codes per attribution; dyads recurring across incidents) and was not treated as duplication.

1.6 Referential Integrity Across Files

Cross-checking incident_id between the global (master) table and the three long-format tables surfaced two distinct patterns:

- Receiver & Attribution: fully consistent with Global other than 92 incidents present in Global but absent from both — the same 92 incidents in each case.
- Dyadic: 111 incident_ids appear in Dyadic but not in Global, and 568 Global incidents are absent from Dyadic.

This is very likely a dataset-versioning artifact — the dyadic file is v0.1 while global/receiver/attribution are v1.3 — rather than a cleaning error, but it means incident-level joins across Dyadic and Global will lose rows on both sides.

1.7 Final Cleaned Shapes

Dataset	Rows	Columns
Global	3,414	84 (was 85)
Receiver	12,180	6 (unchanged)
Attribution	5,188 (was 5,217)	16 (was 18)
Dyadic	4,296	51 (was 57)

2. Exploratory Data Analysis

2.1 Incidents Over Time

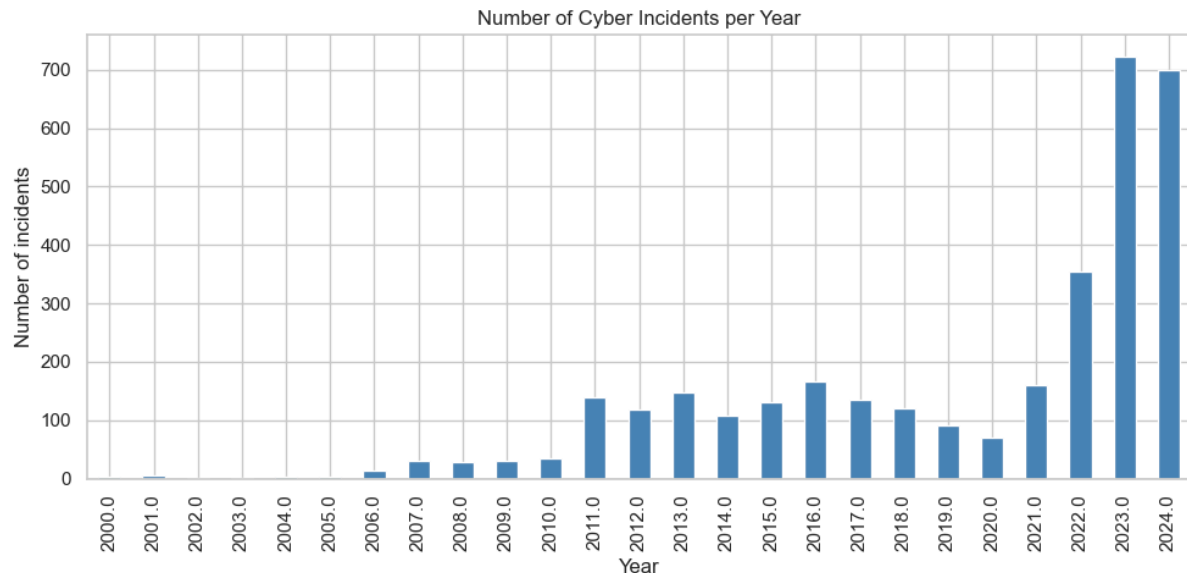


Figure 1. Number of recorded cyber incidents per year.

Incident volume rises sharply through the 2010s and peaks in the most recent, most fully-recorded years, consistent with both a genuine increase in cyber activity and improved contemporary disclosure/tracking. Early years (pre-2010) are sparsely represented, likely reflecting incomplete historical records rather than an actual lull in incidents.

2.2 Top Initiator Countries

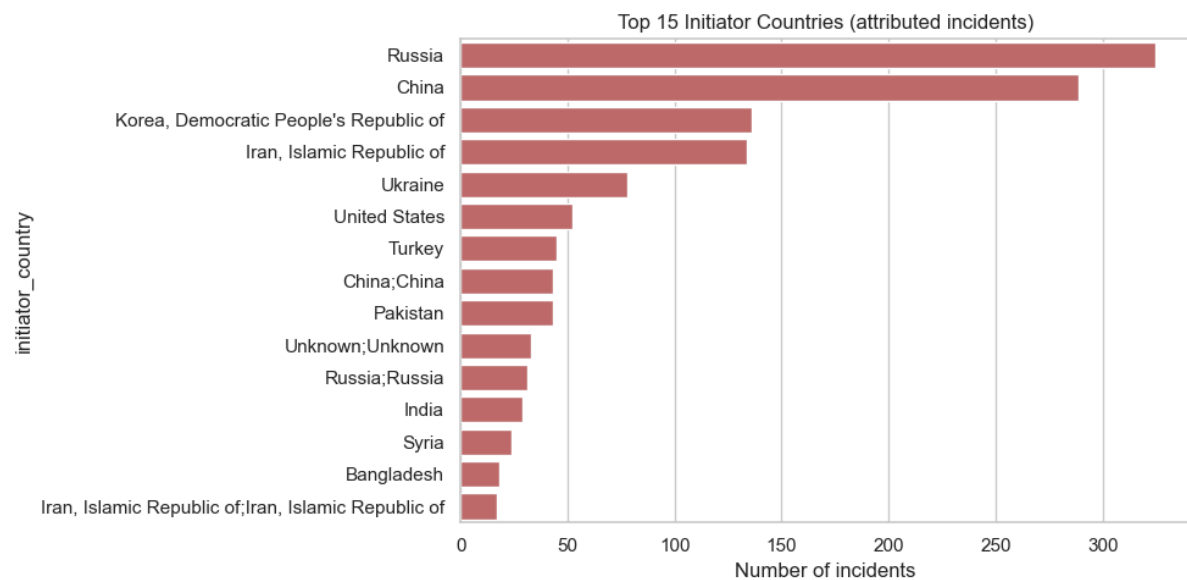


Figure 2. Top 15 attributed initiator countries.

Only 72.7% of incidents in the raw data carry an attributed initiator country; the remaining 27.3% are recorded as "Not attributed." Among incidents that are attributed, a small number of states account for a disproportionate share, consistent with well-documented state-sponsored cyber activity concentrated among a handful of geopolitically active actors.

2.3 Top Receiver (Target) Countries

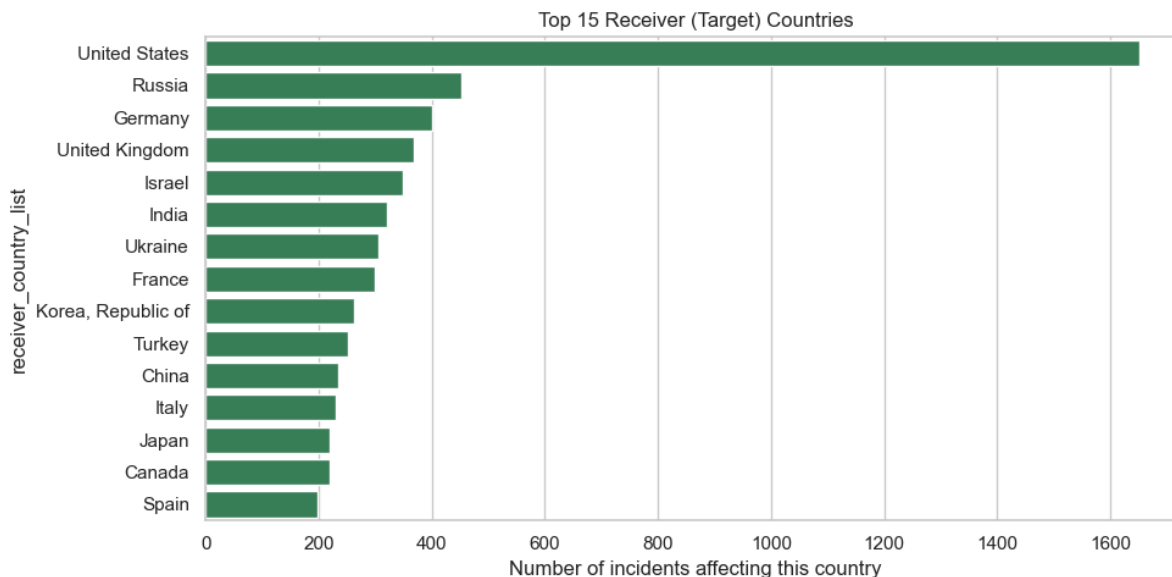


Figure 3. Top 15 receiver (target) countries, counted from the exploded multi-value field.

Because a single incident can affect several countries at once, this chart uses the exploded receiver_country list so that each affected country is counted once per incident regardless of how many countries a given incident touches.

2.4 Incident Type Distribution

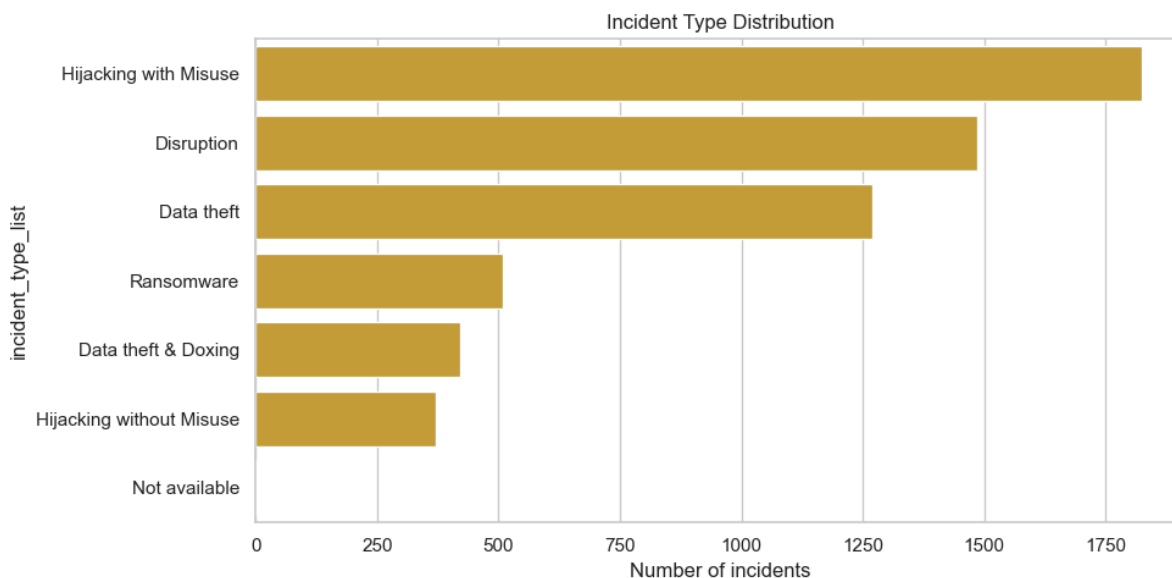


Figure 4. Distribution of incident types (exploded multi-value field).

Data theft and disruption-oriented incident types dominate the record, reflecting both the prevalence of espionage/exfiltration campaigns and the ease with which disruptive attacks (DDoS, defacement) are detected and publicly reported.

2.5 Receiver Category — Who Gets Targeted?

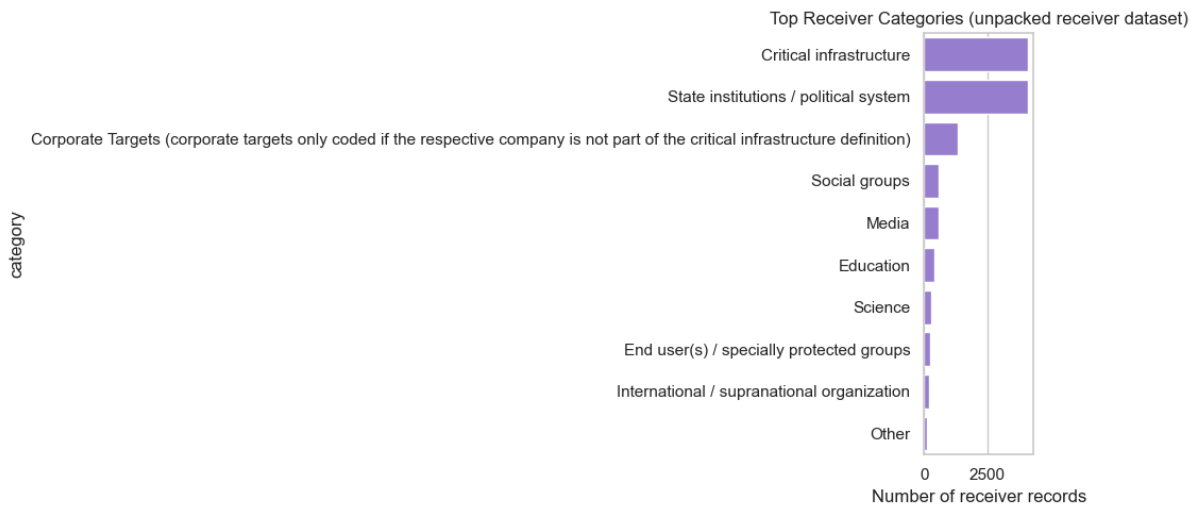


Figure 5. Top receiver categories from the unpacked receiver dataset.

State institutions and critical infrastructure are consistently among the most-targeted categories, underscoring that a large share of tracked cyber incidents have a geopolitical or national-security dimension rather than being purely commercial cybercrime.

2.6 Incident Intensity

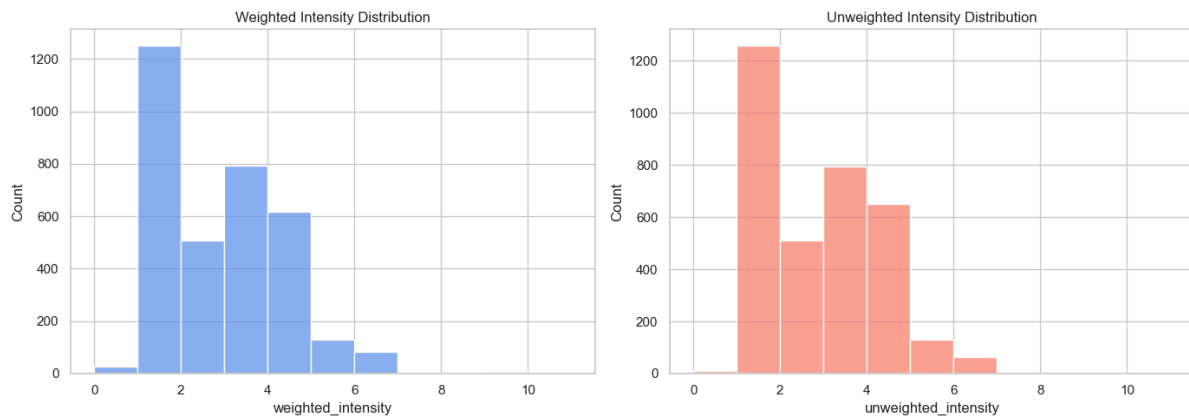


Figure 6. Weighted vs. unweighted intensity score distributions.

Statistic	Weighted intensity	Unweighted intensity
Mean	2.44	2.43
Std. dev.	1.43	1.35
Median	2	2
Max	11	7

Both measures are right-skewed: most incidents cluster at low-to-moderate intensity (median 2), with a long tail of a small number of unusually severe incidents — the weighted measure's higher max (11 vs. 7) reflects its extra multiplier for incidents against very high-importance targets such as critical infrastructure.

2.7 Attribution Basis & Type

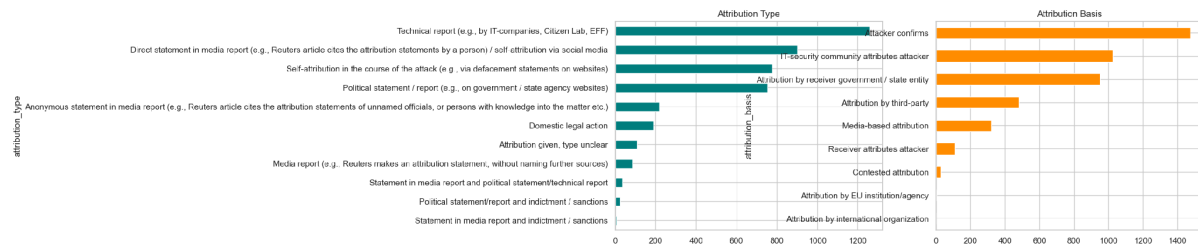


Figure 7. Attribution type (left) and attribution basis (right).

Attribution is most often made via public/media statements and IT-security industry technical reports rather than formal government findings, which is consistent with the broader pattern in cyber-conflict research that firm, state-backed attribution is comparatively rare and slow relative to private-sector reporting.

2.8 Top Initiator → Receiver Country Dyads

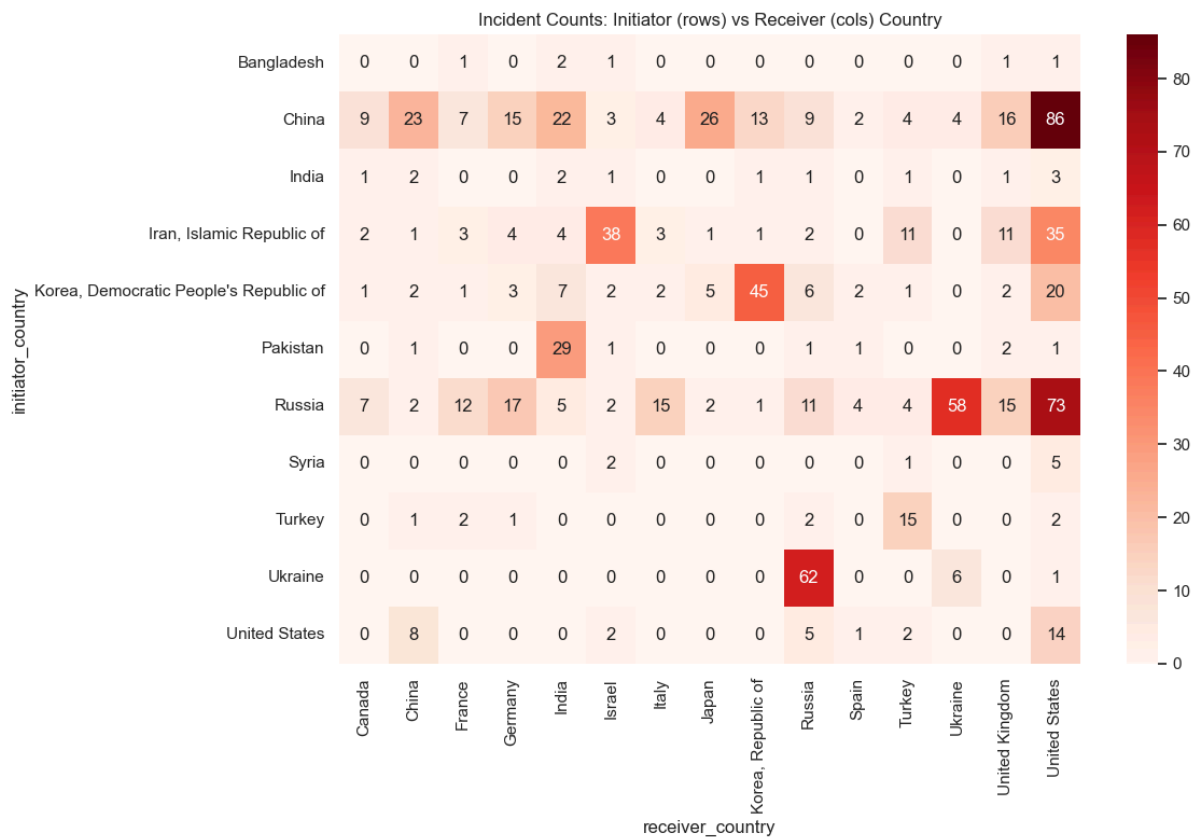


Figure 8. Incident counts by initiator (rows) vs. receiver (columns) country, top countries only.

Initiator	Receiver	Incidents
China	United States	86
Russia	United States	73
Ukraine	Russia	62
Russia	Ukraine	58
North Korea	South Korea	45
Iran	Israel	38
Iran	United States	35

Initiator	Receiver	Incidents
China	Taiwan	31
Pakistan	India	29
China	Japan	26

The dyadic table reflects well-known real-world cyber rivalries — US–China and US–Russia lead in volume, while Russia–Ukraine reflects the ongoing conflict, and North Korea–South Korea and Iran–Israel reflect long-running regional tensions.

2.9 Missing Data After Cleaning

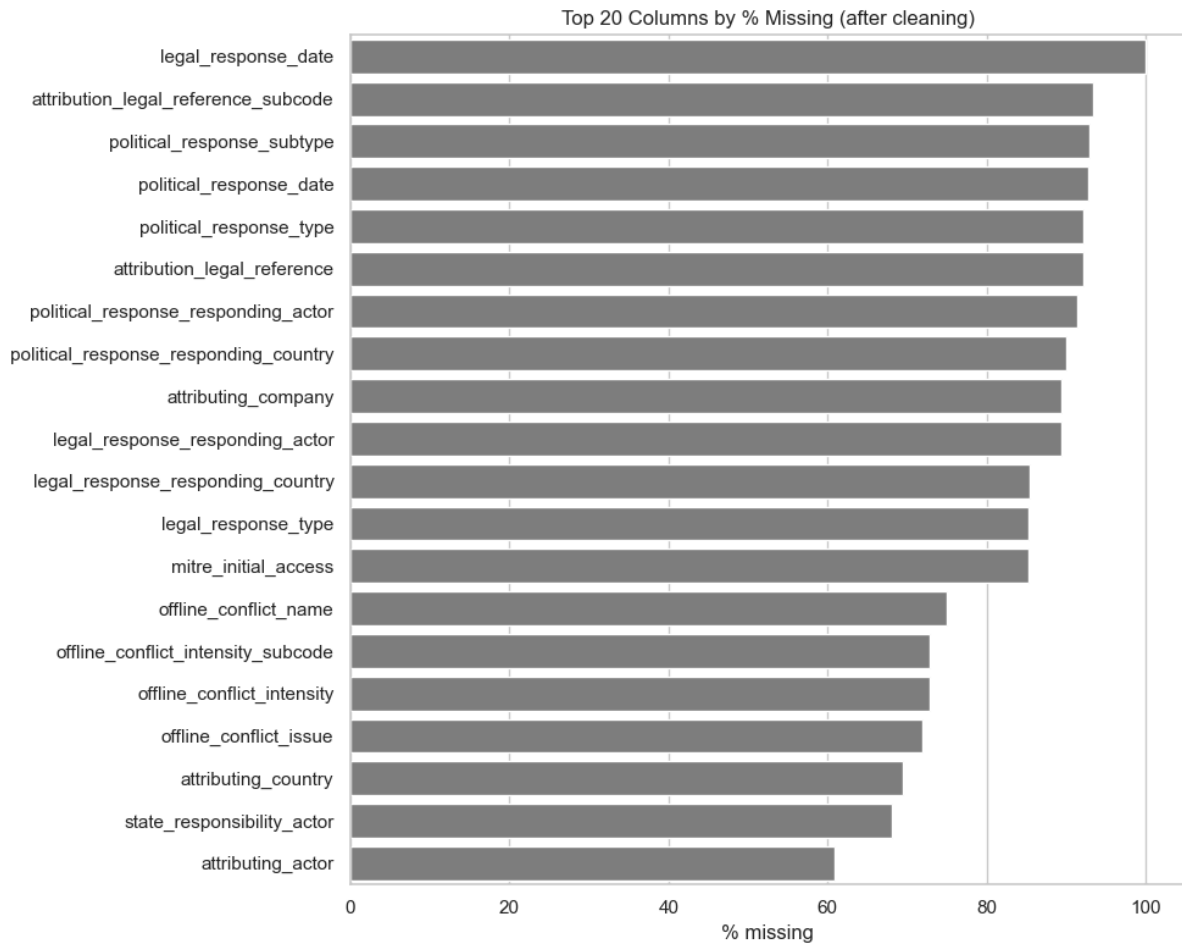


Figure 9. Remaining missingness by column, global dataset, post-cleaning.

Even after standardizing placeholders and dropping the emptiest columns, a number of fields remain heavily incomplete — mostly response/consequence fields (political and legal response details, economic impact) that are inherently only populated when that kind of follow-up actually occurred. This is a structural characteristic of the domain rather than a cleaning gap: most cyber incidents do not receive a formal political or legal response.

2.10 Correlation Between Severity Measures

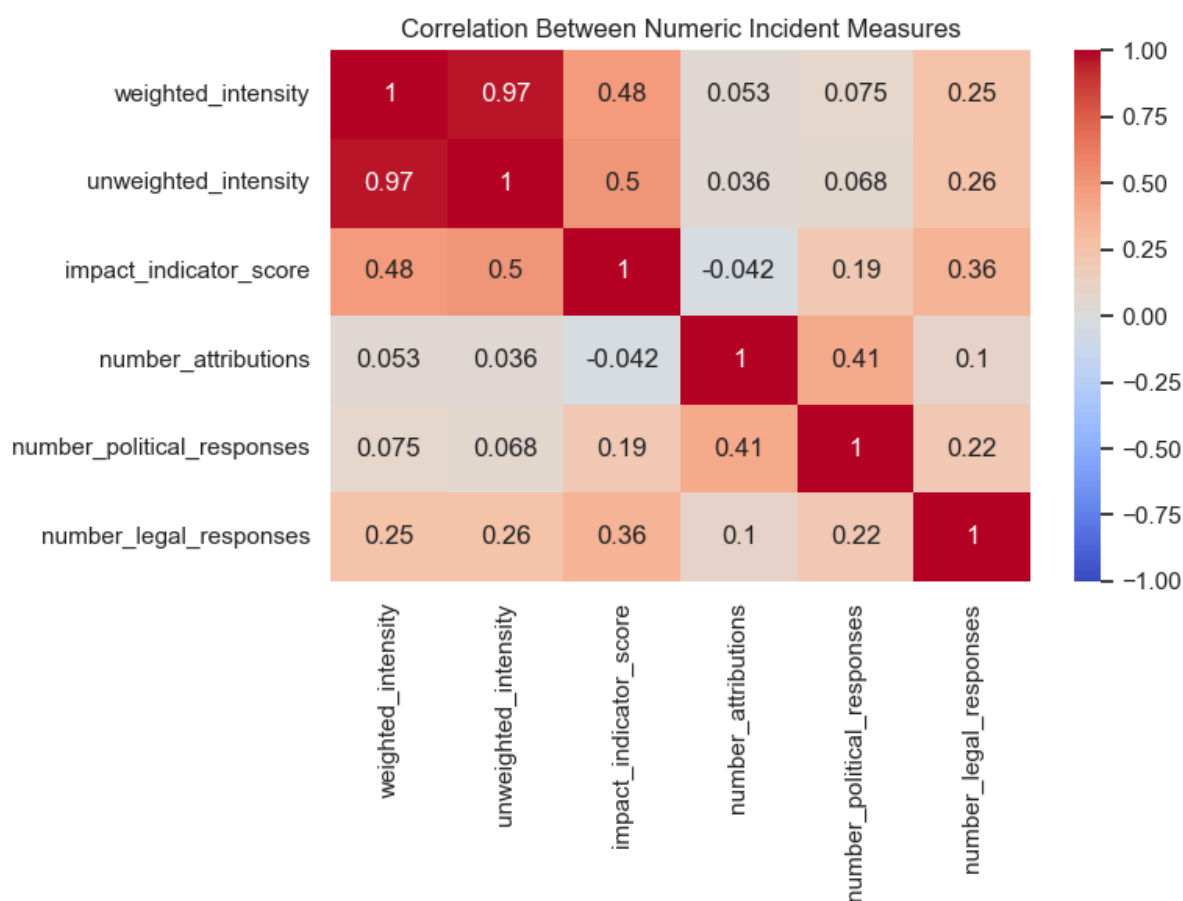


Figure 10. Correlation matrix of numeric severity/response-count measures.

Weighted and unweighted intensity are, unsurprisingly, very strongly correlated with each other. Their correlation with the number of attributions, political responses, and legal responses is comparatively weak, suggesting that how severe an incident is and whether it receives a formal follow-up response are largely independent in this dataset — severity alone does not reliably predict a political or legal reaction.

3. Key Findings

- Incident volume has grown substantially over time, with the sharpest increases in the most recent years covered by the data.
- 72.7% of incidents carry an attributed initiator country; a small set of states (led by China, Russia, Iran, and North Korea in different dyads) account for a disproportionate share of attributed activity.
- State institutions and critical infrastructure are the most frequently targeted receiver categories, reinforcing the dataset's geopolitical/national-security framing.
- Data theft and disruption are the leading incident types by volume.
- Incident intensity is right-skewed — most incidents are low-to-moderate severity, with a small number of high-intensity outliers driven mainly by attacks on critical-infrastructure-class targets.
- Attribution is driven more by media reporting and private security research than by formal government statements.
- The leading country dyads (China–US, Russia–US, Russia–Ukraine, North Korea–South Korea, Iran–Israel) track closely with well-known real-world geopolitical rivalries.
- Incident severity and the likelihood of a formal political/legal response appear to be largely independent.

4. Data Quality Notes & Caveats

- Dyadic dataset is a different version (v0.1) from the other three (v1.3), which explains the `incident_id` mismatches found in §1.6 — not a cleaning defect.
- `settled_attribution` ends up entirely null after the boolean-conversion step due to a dtype mismatch in the mapping logic (§1.4) — flagged for a follow-up fix rather than silently dropped.
- 92 incidents present in the Global dataset have no corresponding Receiver or Attribution records — worth spot-checking whether these are genuinely un-enriched incidents or a data-export gap.
- Several response/consequence fields remain mostly empty even after cleaning; this reflects that most incidents simply don't receive a formal follow-up, not a data-processing shortfall.

Statistical & Distributional Analysis

1. Descriptive Statistics (Numeric Variables)

Nine numeric variables were retained for quantitative summary after excluding identifier and date-like columns. Descriptive statistics (mean, standard deviation, and quartiles) are shown below.

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Number of attributions	1.14	0.51	0	1	1	1	10
Number of political responses	0.11	0.50	0	0	0	0	13
Unweighted intensity	2.43	1.35	0	1	2	3	7
Weighted intensity	2.44	1.43	0	1	2	3	11
Impact indicator score	3.94	3.61	0	0	5	7	15
Affected entities value	297.6	17,115.3	0	0	0	1	1,000,000
Affected third countries value	0.48	0.88	0	0	0	1	23
Economic impact value (\$)	857,210.9	19,455,740	0	0	0	0	872,000,000
Number of legal responses	0.15	0.39	0	0	0	0	3

The large gap between the mean and median for economic_impact_value and affected_entities_value (means far exceeding medians of 0) indicates strong right-skew: most incidents involve little to no measurable economic loss or entity impact, but a small number of extreme incidents pull the mean up dramatically — consistent with a heavy-tailed, "black swan" incident pattern typical of cyber conflict data.

2. Distributions

Histograms with kernel density estimates were used to examine the shape of the core severity and response metrics.

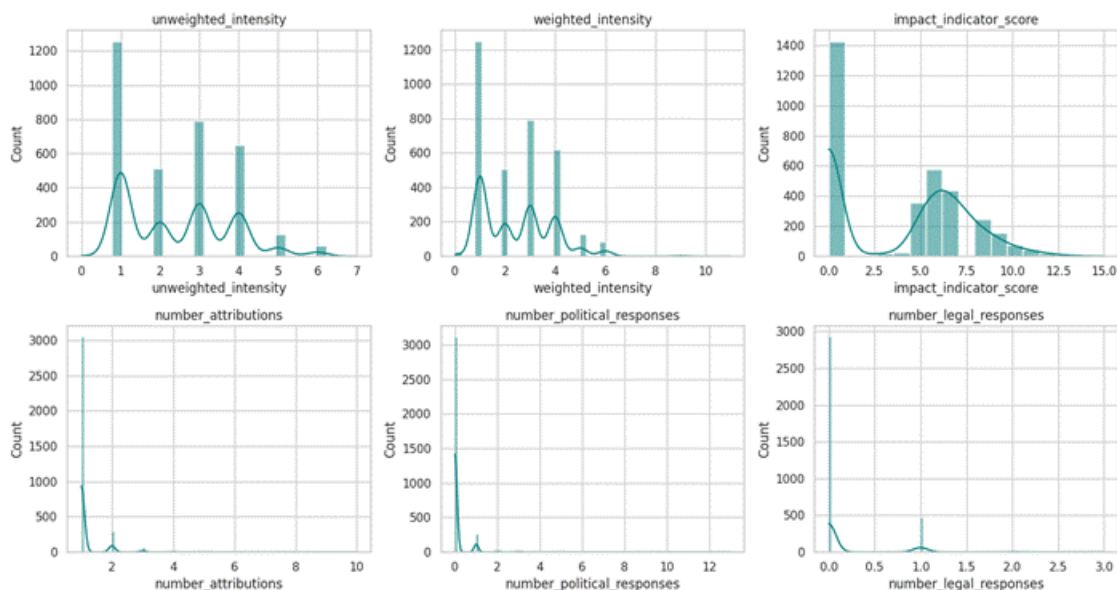


Figure 2. Distributions of intensity, impact, and response-count variables.

Most incidents cluster at low intensity and impact scores, with counts of attributions, political responses, and legal responses concentrated near zero — the large majority of incidents receive a single attribution and no formal political or legal response.

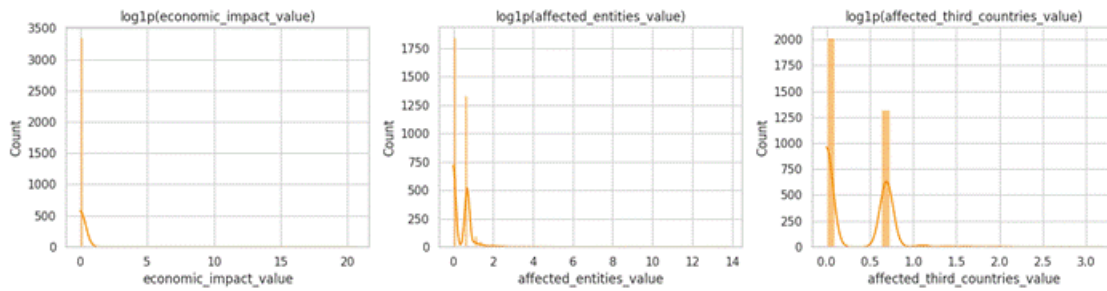


Figure 3. *Log1p-transformed distributions of economic impact, affected entities, and affected third countries.*

Because `economic_impact_value` and `affected_entities_value` are extremely right-skewed in raw form, a `log1p` transform was applied for visualization. This reveals that even after log-scaling, the bulk of incidents remain concentrated at the low end, with a long tail of substantially larger-scale events.

3. Outlier Detection

Outliers were identified using two complementary methods: the $1.5 \times \text{IQR}$ rule and a Z-score threshold of $|Z| > 3$. Boxplots were also used for visual inspection.

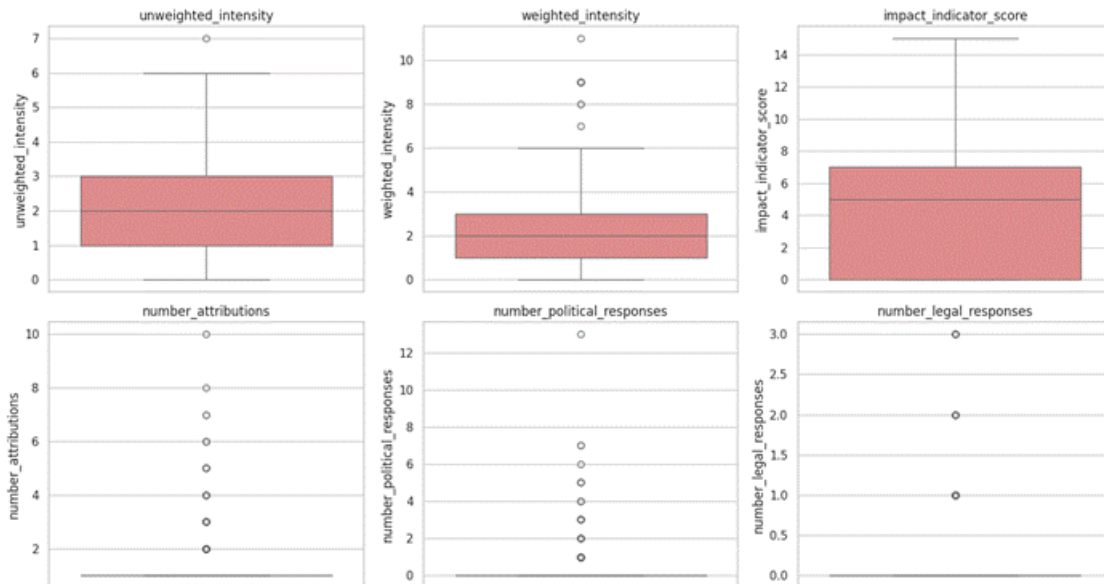


Figure 4. *Boxplots of intensity, impact, and response-count variables.*

In this dataset, boxplot outliers largely represent unusually severe or high-response incidents — the statistical equivalent of "black swan" cyber events that are meaningfully more severe or drew significantly more attributions/responses than a typical incident, rather than data errors.

3.1 IQR-Based Outlier Counts

Variable	Outliers (IQR rule)	% of Records	Lower Bound	Upper Bound
Number of legal responses	488	14.29%	0.0	0.0
Number of attributions	366	10.72%	1.0	1.0
Number of political responses	297	8.70%	0.0	0.0
Affected entities value	151	4.42%	-1.5	2.5
Economic impact value	70	2.05%	0.0	0.0
Affected third countries value	45	1.32%	-1.5	2.5

Weighted intensity	11	0.32%	-2.0	6.0
Unweighted intensity	1	0.03%	-2.0	6.0
Impact indicator score	0	0.00%	-10.5	17.5

3.2 Z-Score-Based Outlier Counts ($|Z| > 3$)

Variable	Outliers ($ Z > 3$)
Number of attributions	72
Number of political responses	40
Affected third countries value	32
Number of legal responses	28
Weighted intensity	11
Economic impact value	9
Impact indicator score	1
Unweighted intensity	1
Affected entities value	1

number_legal_responses (14.3%) and number_attributions (10.7%) show the highest outlier rates under the IQR rule, reflecting that while most incidents receive exactly one attribution and no legal response, a meaningful minority involve multiple attributions or legal actions. impact_indicator_score shows zero IQR outliers, indicating a comparatively well-behaved, non-extreme distribution relative to the other metrics.

4. Relationships Between Variables

4.1 Scatterplots

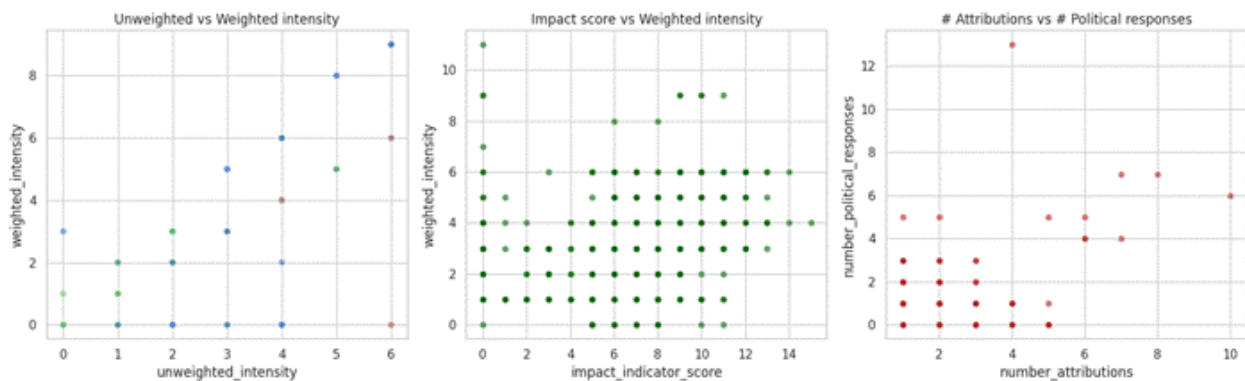


Figure 5. Pairwise relationships between intensity, impact, and response metrics.

Unweighted and weighted intensity move together almost linearly, as expected since weighted intensity is a refinement of the unweighted measure. The relationship between number of attributions and number of political responses is examined to assess whether states respond more assertively when attacker attribution is more certain.

4.2 Pairplot by Initiator Category

Figure 7. Weighted intensity by initiator category.

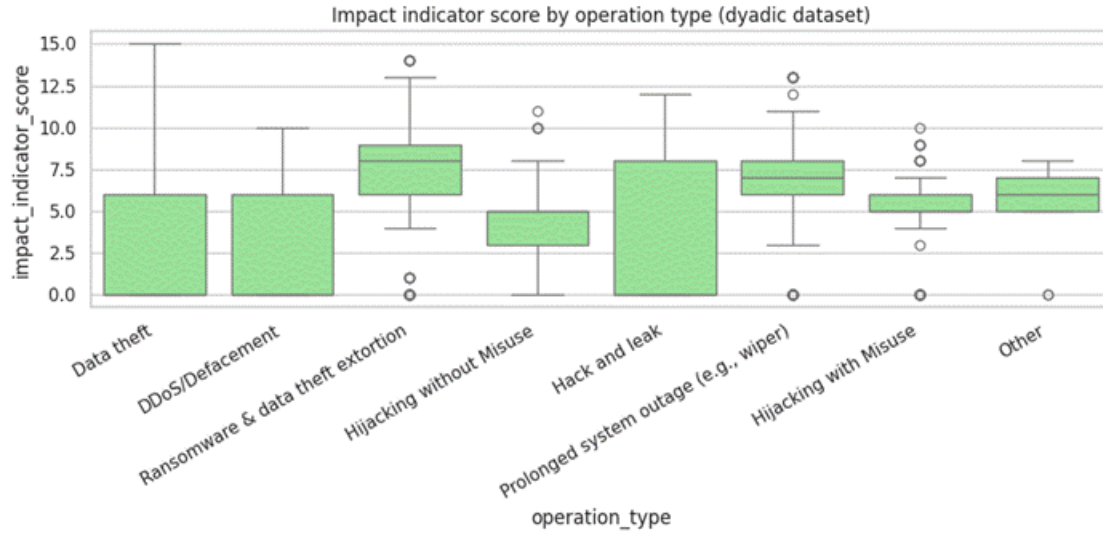


Figure 8. Impact indicator score by operation type.

Ransomware and data-theft/extortion operations tend to show a higher median impact indicator score than simpler operation types such as basic data theft, consistent with the disruptive and coercive nature of ransomware and extortion campaigns.

5.1 One-Way ANOVA: Impact by Initiator Category

A one-way ANOVA was used to formally test whether `impact_indicator_score` differs significantly across `initiator_category` groups.

Test	F-statistic	p-value	Conclusion
One-way ANOVA (<code>impact_indicator_score ~ initiator_category</code>)	54.415	< 0.0001	Statistically significant difference between groups

With $F = 54.415$ and $p < 0.0001$, the null hypothesis of equal mean impact scores across initiator categories is rejected at any conventional significance level. This provides strong statistical evidence that the type of initiating actor (e.g., state-sponsored vs. non-state) is associated with meaningfully different incident impact levels.

6. Key Findings & Summary

- **Distributional shape:** Severity and response metrics (intensity, impact, attribution/response counts) are concentrated at low values, with a small number of high-severity "black swan" incidents driving the tails of the distributions.
- **Skewed impact:** `economic_impact_value` and `affected_entities_value` are extremely right-skewed even after log-transformation, indicating that a small number of incidents account for a disproportionate share of total economic and entity-level impact.
- **Outliers:** `number_legal_responses` and `number_attributions` have the highest outlier rates (14.3% and 10.7% under the IQR rule), while `impact_indicator_score` shows no IQR-flagged outliers, suggesting it is a comparatively stable, well-behaved composite measure.
- **Strong relationship:** Unweighted and weighted intensity are almost perfectly correlated, as expected given their construction.

- **Operation type matters:** Ransomware and data-theft/extortion operations are associated with higher impact indicator scores than simpler operation types.
- **Initiator category is significant:** Impact indicator score differs significantly across initiator categories (one-way ANOVA, $F = 54.415$, $p < 0.0001$), indicating that who initiates an incident is statistically associated with how severe its impact is.
- **Data quality caveat:** High missingness in fields like `legal_response_date` most likely reflects the conditional structure of the data (the underlying event did not occur) rather than data-quality issues, and should be handled accordingly in downstream modeling.

Overall, the exploratory analysis indicates that cyber conflict incidents follow a heavy-tailed severity distribution dominated by many low-impact events and a minority of extreme incidents, and that both the actor initiating an incident and the type of operation used are meaningfully associated with its resulting impact.